Arjuna JEE 2027

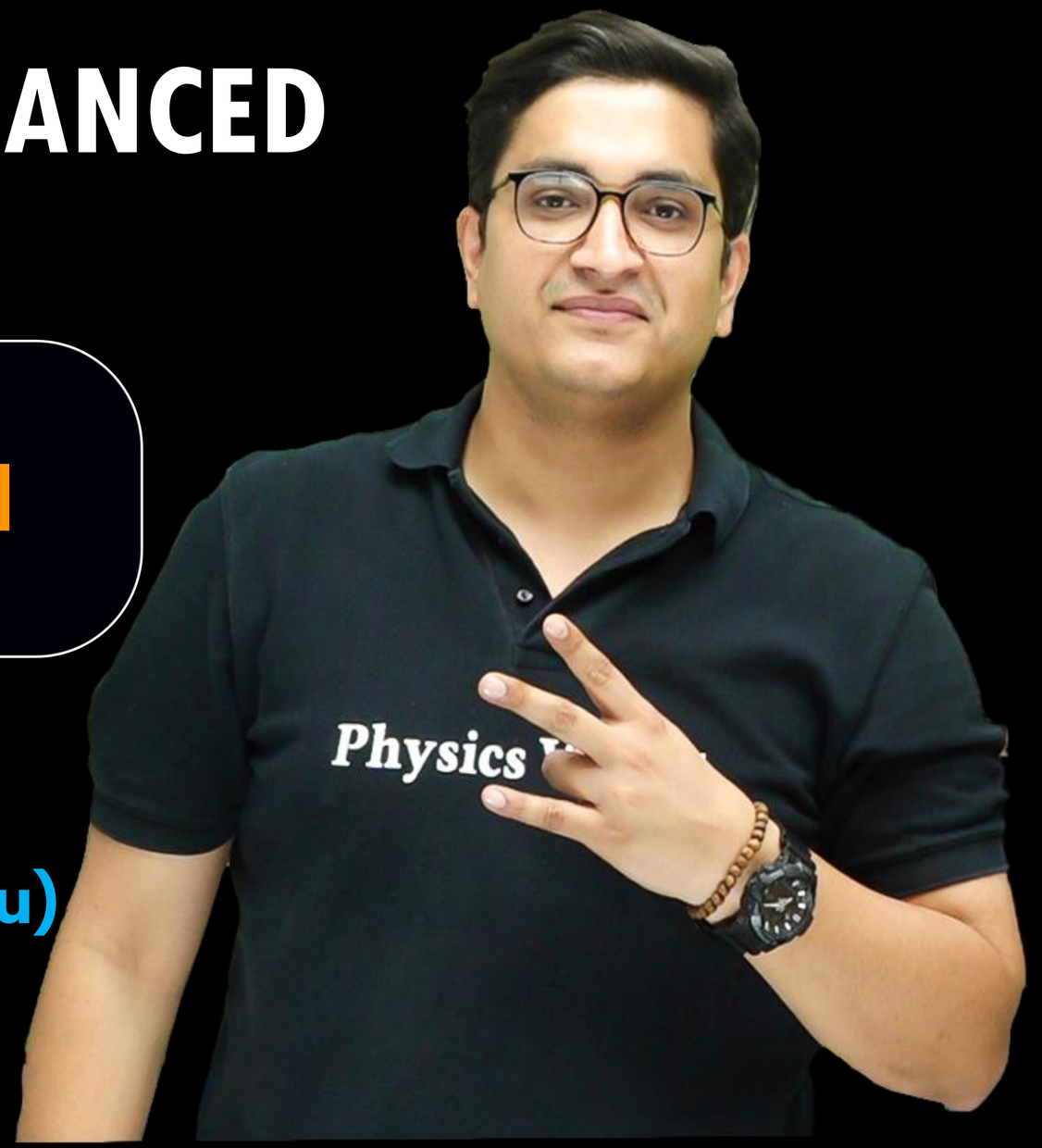
JEE MAINS & IIT JEE ADVANCED

MATHEMATICS

QUADRATIC EQUATION

DIBY (DO IT BY YOURSELF)

By: Sachin Jakhar (Defender Mamu)



DIBY-01

The quadratic equation whose roots are $1+\sqrt{2}, 1-\sqrt{2}$ is

(A) $x^2+2x-1=0$ **(B)** $x^2-2x-1=0$

(C) $x^2-2x+1=0$ **(D)** $x^2+2x+1=0$

[Ans. : (B)]

DIBY-02

If α, β are roots of the equation $ax^2-bx-c=0$, then $\alpha^2-\alpha\beta+\beta^2$ is equal to

(A) $\frac{b^2+3ac}{a^2}$ **(B)** $\frac{b^2-3ac}{a^2}$ **(C)** $\frac{b^2+2ac}{a^2}$ **(D)** $\frac{b^2-2ac}{a^2}$

[Ans. : (A)]

DIBY-03

If α, β are the roots of $ax^2+bx+c=0$, then the equation whose roots are $2+\alpha$ and $2+\beta$ is

(A) $ax^2+(b-4a)x+(4a-2b+c)=0$ **(B)** $ax^2-(b-4a)x+(4a-2b+c)=0$

(C) $ax^2+(b-4a)x-(4a-2b+c)=0$ **(D)** $ax^2-(b-4a)x-(4a-2b+c)=0$

[Ans. : (A)]

DIBY-04

If α and β are roots of the equation $x^2 + px + \frac{3p}{4} = 0$, such that $|\alpha - \beta| = \sqrt{10}$, then p belongs to the set

(A) $\{2, -5\}$ (B) $\{-3, 2\}$ (C) $\{3, -5\}$ (D) $\{-2, 5\}$

[Ans. : (D)]

DIBY-05

If α, β are the roots of $ax^2 + bx + c = 0$ and $c \neq 0$ then the value of $\frac{1}{(a\alpha+b)^2} + \frac{1}{(a\beta+b)^2}$ in terms of a, b, c is

(A) $\frac{b^2-2ac}{ac}$ (B) $\frac{2ac-b^2}{ac}$ (C) $\frac{b^2-2ac}{a^2c^2}$ (D) $\frac{b^2}{2}$

[Ans. : (C)]

DIBY-06

Let α and β be the roots of the quadratic equation $(x-2)(x-3) + (x-3)(x+1) + (x+1)(x-2) = 0$. Find the value of $\frac{1}{(\alpha+1)(\beta+1)} + \frac{1}{(\alpha-2)(\beta-2)} + \frac{1}{(\alpha-3)(\beta-3)}$

[Ans. : 0]

DIBY-07

If α, β are the roots of $ax^2 + bx + c = 0, (a \neq 0)$ and $\alpha + \delta, \beta + \delta$ are the roots of $Ax^2 + Bx + C = 0, (A \neq 0)$ for some constant δ, then prove that, $\frac{b^2-4ac}{a^2} = \frac{B^2-4AC}{A^2}$.

[JEE-2000]

DIBY-08

If α and β be two roots of the equation $x^2 - 64x + 256 = 0$. Then the value of $\left(\frac{\alpha^3}{\beta^5}\right)^{\frac{1}{8}} + \left(\frac{\beta^3}{\alpha^5}\right)^{\frac{1}{8}}$ is

(A) 1 (B) 3 (C) 4 (D) 2

[Ans. : (D)]

DIBY-09

Let α, β be the roots of the equation $x^2 + 2\sqrt{2}x - 1 = 0$. The quadratic equation, whose roots are $\alpha^4 + \beta^4$ and $\frac{1}{10}\left(\alpha^6 + \beta^6\right)$, is: [JEE Mains-2024]

(A) $x^2 - 190x + 9466 = 0$
(B) $x^2 - 195x + 9466 = 0$
(C) $x^2 - 195x + 9506 = 0$
(D) $x^2 - 180x + 9506 = 0$

[Ans. : (C)]

DIBY-10

If the equation $a(b-c)x^2+b(c-a)x+c(a-b)=0$ has equal roots, where $a+c=15$ and $b=\frac{36}{5}$, then a^2+c^2 is equal to **[JEE Mains-2025]**

[Ans. : 117]

DIBY-11

Let $\alpha \in \mathbb{R}$ and let α, β be the roots of the equation $x^2+60^{\frac{1}{4}}x+a=0$.
If $\alpha^4+\beta^4=-30$, then the product of all possible values of a is **[JEE Mains-2023]**

[Ans. : 45]

DIBY-12

If 2 and 6 are the roots of the equation $ax^2+bx+1=0$, then the quadratic equation, whose roots are $\frac{1}{2a+b}$ and $\frac{1}{6a+b}$, is : **[JEE Mains-2024]**

(A) $x^2+8x+12=0$ (B) $2x^2+11x+12=0$

(C) $4x^2+14x+12=0$ (D) $x^2+10x+16=0$

[Ans. : (A)]

DIBY-13

Let α and β be the roots of the equation, $5x^2 + 6x - 2 = 0$. If $S_n = \alpha^n + \beta^n, n = 1, 2, 3 \ldots.$, then [JEE Mains-2020]

(A) $6S_6 + 5S_5 = 2S_4$
(B) $6S_6 + 5S_5 + 2S_4 = 0$
(C) $5S_6 + 6S_5 = 2S_4$
(D) $6S_6 + 6S_5 + 2S_4 = 0$

[Ans. : (C)]

DIBY-14

If α, β are the roots of the equation, $x^2 - x - 1 = 0$ and $S_n = 2023\alpha^n + 2024\beta^n$, then [JEE Mains-2024]

(A) $2S_{12} = S_{11} + S_{10}$
(B) $S_{12} = S_{11} + S_{10}$
(C) $2S_{11} = S_{12} + S_{10}$
(D) $S_{11} = S_{10} + S_{12}$

[Ans. : (B)]

DIBY-15

Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to [JEE Mains-2024]

[Ans. : 4]

DIBY-16

Let $\alpha, \beta; \alpha > \beta$, be the roots of the equation $x^2 - \sqrt{2}x - \sqrt{3} = 0$. Let $P_n = \alpha^n - \beta^n, n \in N$. Then $(11\sqrt{3} - 10\sqrt{2})P_{10} + (11\sqrt{2} + 10)P_{11} - 11P_{12}$ is equal to:

[JEE Mains-2024]

(A) $10\sqrt{2}P_9$ (B) $10\sqrt{3}P_9$ (C) $11\sqrt{2}P_9$ (D) $11\sqrt{3}P_9$

[Ans. : (B)]

DIBY-17

Let α and β be the roots of $x^2 - 6x - 2 = 0$. If $a_n = \alpha^n - \beta^n$ for $n \geq 1$, then the value of $\frac{a_{10} - 2a_8}{3a_9}$ is:

[JEE Mains-2021]

(A) 2 (B) 1 (c) 4 (d) 3

[Ans. : (A)]

DIBY-18

Let $\alpha, \beta(\alpha > \beta)$ be the roots of the quadratic equation $x^2 - x - 4 = 0$. If $P_n = \alpha^n - \beta^n, n \in N$, then $\frac{P_{15}P_{16} - P_{14}P_{16} - P_{15}^2 + P_{14}P_{15}}{P_{13}P_{14}}$ is equal to.

[JEE Mains-2022]

[Ans. : 16]

DIBY-19

The number of real solutions of the equation, $x^2 - |x| - 12 = 0$ is:

[JEE Mains-2021]

(A) 1 **(B)** 4 **(C)** 3 **(D)** 2

[Ans. : (D)]

DIBY-20

The sum, of the squares of all the roots of the equation $x^2 + |2x - 3| - 4 = 0$, is

[JEE Mains-2025]

(A) $6(2 - \sqrt{2})$ **(B)** $3(3 - \sqrt{2})$ **(C)** $6(3 - \sqrt{2})$ **(D)** $3(2 - \sqrt{2})$

[Ans. : (A)]

DIBY-21

The product of the roots of the equation $9x^2 - 18|x| + 5 = 0$, is:

[JEE Mains-2020]

(A) $\frac{25}{9}$ **(B)** $\frac{25}{81}$ **(C)** $\frac{5}{9}$ **(d)** $\frac{5}{27}$

[Ans. : (B)]

QUADRATIC INVOLVING MODULUS

DIBY-22

The sum of the squares of the roots of $|x+2|^2+|x-2|-2=0$ and the squares of the roots of $x^2-2|x-3|-5=0$, is **[JEE Mains-2025]**

(A) 26 (B) 36 (C) 30 (d) 24

[Ans. : (B)]

DIBY-23

The number of real roots of the equation $x|x|-5|x+2|+6=0$

[JEE Mains-2023]

(A) 5 (B) 3 (C) 6 (D) 4

[Ans. : (B)]

DIBY-24

The sum of all the roots of the equation $\left|x^2-8x+15\right|-2x+7=0$ is:

[JEE Mains-2023]

(A) $9+\sqrt{3}$ (B) $11+\sqrt{3}$ (C) $9-\sqrt{3}$ (D) $11-\sqrt{3}$

[Ans. : (A)]

DIBY-25

The number of real solutions of the equation $x|x+5|+2|x+7|-2=0$ is

[JEE Mains-2024]

[Ans. : 3]

DIBY-26

If p and q are non-zero real numbers and $\alpha^3+\beta^3=-p, \alpha\beta=q$, then a quadratic equation whose roots are $\frac{\alpha^2}{\beta}, \frac{\beta^2}{\alpha}$ is

(A) $qx^2+px+q^2=0$

(B) $px^2+qx+p^2=0$

(C) $qx^2-px+q^2=0$

(D) $px^2-qx+p^2=0$

[Ans. : (A)]

DIBY-27-A(Challenger)

Let $f(x)=ax^2+bx+c, a, b, c \in I$ & $f(1)=0, 50<f(7)<60$ & $70<f(8)<80$, then $f(-1)$ is equal to

[Ans. : 14]

DIBY-27-B

If roots of the equation $(x-\alpha)(x-\beta)+(x-\gamma)(x-\delta)=0$ are m & n respectively, then roots of the equation $2(x-m)(x-n)-(x-\alpha)(x-\beta)=0$ Will be

(A) $\frac{\alpha+m+\gamma}{2}, \frac{\beta+n+\delta}{2}$ **(B)** γ, δ

(C) $\alpha-m, \beta-n$ **(D)** $m-\alpha+\gamma, n-\beta+\delta$

[Ans. : (B)]

DIBY-28

If the roots of the equation $ax^2+x+b=0$ be real and different, then the roots of the equation $x^2-4\sqrt{ab}x+1=0$ will be

(A) Rational **(B)** Irrational

(C) Real **(D)** Imaginary **[Ans. : (D)]**

DIBY-29

Let α, β are the roots of the equation $x^2+ax+b=0$ such that $\frac{3\alpha-5-\alpha^2}{(\beta-3)}=\beta$, then value of $|a^2-2b+3a|$ is (where, $\beta \neq 3$)

(A) 0 **(B)** 3 **(C)** 5 **(D)** 7

[Ans. : (C)]

DIBY-30

If α is a root of the equation $4x^2 - 2x + 3 = 0$, then its other roots is

(A) $\frac{2}{3} + \alpha$ (B) $\frac{\alpha+1}{2\alpha-1}$

(C) $\frac{3}{2(4\alpha^2+3)}$ (D) $\frac{3\alpha}{4}$

[Ans. : B, C]

DIBY-31

If roots of $2(x-a)(x-b)+c=0$ are α, β then roots $(x-\alpha)(x-\beta)-\frac{c}{2}=0$ are

(A) lies between a & b (B) $-a, -b$

(C) a, b (D) $2a, 2b$

[Ans. : (C)]

DIBY-32

If the equation in x, $x^2 + 2(1+\lambda)x + (3\lambda^2+2) = 0$ has imaginary roots, then

(A) $\lambda = 1$ (B) $\lambda = -\frac{1}{2}$ (C) $\lambda = 0$ (D) $\lambda = -1$

[Ans. : A, B, C, D]

DIBY-33

If the equation $\frac{x^2-bx}{ax-c}=\frac{m-1}{m+1}$ has roots equal in magnitude but opposite in sign, then m is

(A) $\frac{a+b}{a-b}$ (B) $\frac{a-b}{a+b}$ (C) $\frac{b-a}{b+a}$ (D) None of these

[Ans. : (B)]

DIBY-34

Find the value of a for which one root of the equation $x^2+(2a-1)x+a^2+2=0$ is twice as large as the other.

[Ans. : $a=-4$]

DIBY-35

If $x=3+\sqrt{5}$ then the value of $x^4-12x^3+44x^2-48x+17$.

(A) –1 (B) 0 (C) 1 (D) 2

[Ans. (C)]

DIBY-36

The quadratic equation $x^2 + ax + b + 1 = 0$ has both positive integral roots, then $a^2 + b^2$ can be equal to

(A) 50 (B) 37 (C) 61 (D) 20

[Ans. : A, D]

DIBY-37

The number of real roots of the equation $5 + |2^x - 1| = 2^x(2^x - 2)$ is

[JEE Mains-2019]

(A) 4 (B) 3 (C) 2 (D) 1

[Ans. : (D)]

DIBY-38

Find the product of the real roots of the equation, $x^2 + 18x + 30 = 2\sqrt{x^2 + 18x + 45}$

[Ans. : 20]

DIBY-39

If 'α' is an irrational solution of the equation $3^{2x^2-7x+3} = 4^{x^2-x-6}$, then the value of $3^{\left(\frac{2\alpha-1}{\alpha+2}\right)}$ is

[Ans. : 4]

GRAPH OF QUADRATIC

DIBY-40

Consider the graph $f(x) = ax^2 + bx + c$ and comment on sign of following parts:

(i) (abc) **is** _____.

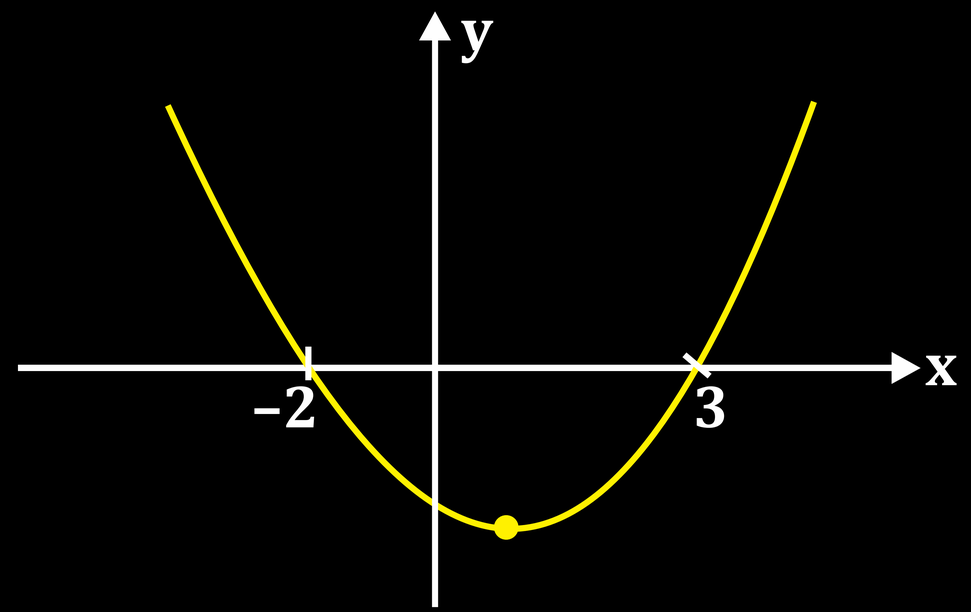
(ii) $(4a + 2b + c)$ **is** _____.

(iii) for all $x > 3, f(x)$ **is** _____.

(iv) For all $-2 < x < 3, f(x)$ **is** _____.

(v) D **is** _____.

(vi) $(a - b + c)\cdot(9a - 3b + c)$ **is** _____.



Ans. :

(i) +ve, (ii) –ve, (iii) +ve, (iv) –ve, (v) +ve, (vi) –ve

DIBY-41(Challenger)

Draw the graph of quadratic $(y = ax^2 + bx + c)$ in each part with given conditions:

(i) Roots are 3 & 7 and crosses y-axis at (0, –2)

(ii) Imaginary roots with vertex at $\left(-\frac{3}{4}, \frac{1}{2}\right)$ and $a - b + c = 2$.

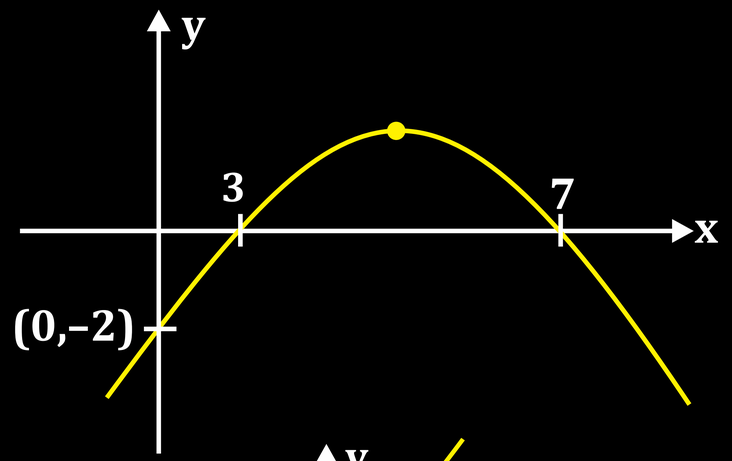
(iii) Touches x-axis at $x = -3$ and value at $x = 2$ is –8.

Also find value of a, b & c in each case.

GRAPH OF QUADRATIC

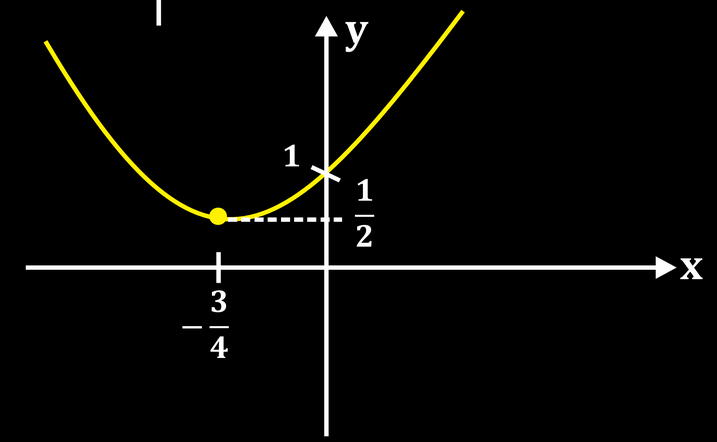
DIBY-41(Answer)

(i)



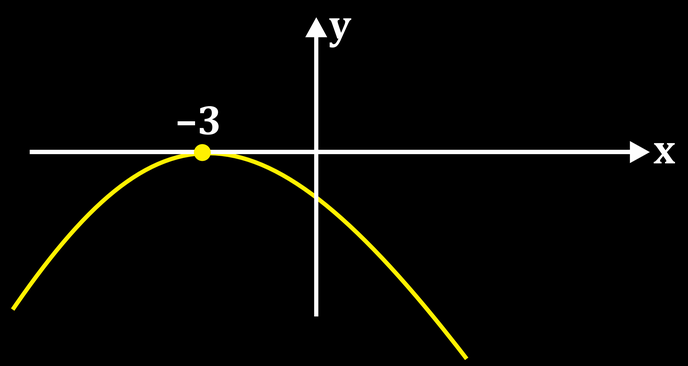
$$a = -\frac{2}{21}, b = \frac{20}{21}, c = -2$$

(ii)



$$a = 24, b = 36, c = 14$$

(iii)



$$a = \frac{-8}{25}, b = \frac{-48}{25}, c = \frac{-72}{25}$$

DIBY-42

If the equation $ax^2 + bx + c = 0, (a > 0)$ has two real roots α and β such that $\alpha < -2$ and $\beta > 2$, then which of the following statements is/are true?

(A) $a - |b| + c < 0$

(B) $c < 0,\ b^2 - 4ac > 0$

(C) $4a - 2|b| + c < 0$

(D) $9a - 3|b| + c < 0$

[Ans. : A, B, C]

DIBY-43

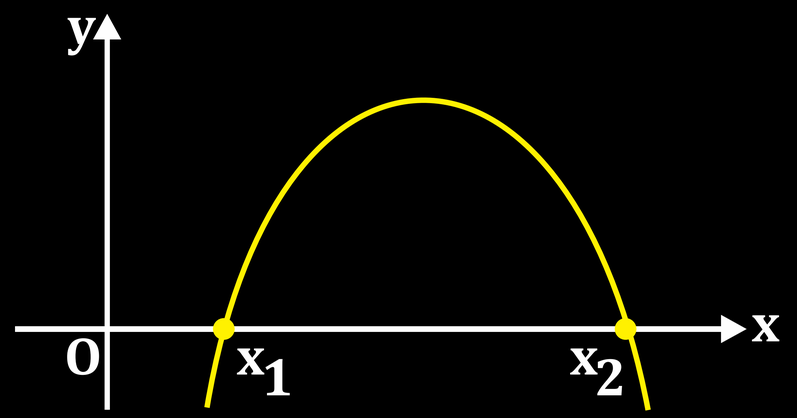
The adjoining figure shows the graph of $y = ax^2 + bx + c$. Then

(A) $a < 0$

(B) $b^2 < 4ax$

(C) $c > 0$

(D) a **and** b **are of opposite sign**



[Ans. : A, D]

RANGE OF QUADRATIC

DIBY-44

Find the range of $f(x)$:

(i) $f(x) = 2x^2 - 3x + 2$

(ii) $f(x) = 2x^2 - 3x + 2, x \in [0, 2]$

[Ans. : (i) $\left[\frac{7}{8}, \infty\right)$, (ii) $\left[\frac{7}{8}, 4\right]$]

DIBY-45

Find range of following functions:

(i) $f(x) = 3x^2 - 2x - 7$

(ii) $f(x) = 3x^2 - 2x - 7, x \in (0, 5]$

(iii) $f(x) = 3x^2 - 2x - 7, x \in [-6, -1]$

[Ans. : (i) $\left[-\frac{22}{3}, \infty\right)$, (ii) $[-22/3, 58]$, (iii) $[-2, 113]$]

RANGE OF RATIONAL FUNCTION

DIBY-46

If the range of the function $f(x) = \frac{5-x}{x^2-3x+2}, x \neq 1, 2$, is $(-\infty, \alpha] \cup [\beta, \infty)$, then $\alpha^2 + \beta^2$ is equal to:

[JEE Mains-2025]

(A) 190 (B) 192 (C) 188 (D) 194

[Ans. : (D)]

DIBY-47

Let $f: \mathbb{R} - \{2, 6\} \to \mathbb{R}$ be real valued function defined as $f(x) = \frac{x^2+2x+1}{x^2-8x+12}$. Then range of f is

[JEE Mains-2023]

(A) $\left(-\infty, -\frac{21}{4}\right] \cup [0, \infty)$

(B) $\left(-\infty, -\frac{21}{4}\right) \cup (0, \infty)$

(c) $\left(-\infty, -\frac{21}{4}\right] \cup \left[\frac{21}{4}, \infty\right)$

(D) $\left(-\infty, -\frac{21}{4}\right] \cup [1, \infty)$

[Ans. : (A)]

DIBY-48

Let the sum of the maximum and the minimum values of the function $f(x) = \frac{2x^2-3x+8}{2x^2+3x+8}$ be $\frac{m}{n}$, where $\gcd(m,n) = 1$. Then $m+n$ is equal to

[JEE Mains-2024]

(A) 195 (B) 201 (C) 217 (D) 182

[Ans. : (B)]

DIBY-49

Show that if x is real then $\frac{x^2-3x+4}{x^2+3x+4}$ lies from 1/7 and 7.

DIBY-50

Find the range of : $f(x) = \frac{x^2+x+2}{x^2+x+1}, x \in R$

[IITJEE-2003]

[Ans. : $\left(1, \frac{7}{3}\right]$]

RANGE OF RATIONAL FUNCTION

DIBY-51

We call 'p' a good number if the inequality $\frac{2x^2+2x+3}{x^2+x+1} \leq p$ is satisfied for any real x. Find the smallest integral good number.

[Ans. : 4]

DIBY-52

If α, β are the roots of equation $x^2 + x - 2 = 0$, then find the value of

$$\frac{\alpha\beta^4(\beta+1)^4+\beta\alpha^4(\alpha+1)^4}{\alpha^2+\beta^2+\alpha+\beta}.$$

[Ans. : -4]

DIBY-53

The minimum value of the sum of the squares of the roots of $x^2 + (3 - a)x + 1 = 2a$ is:

[Ans. : 6]

DIBY-54

The number of real solutions of the equation $3\left(x^2+\frac{1}{x^2}\right)-2\left(x+\frac{1}{x}\right)+5=0$, is

[JEE Mains-2023]

(A) 4 (B) 0 (C) 3 (D) 2

Ans. : (B)

COMMON ROOT CONDITION

DIBY-55

If $x^2+4ax+3=0$ and $2x^2+3ax-9=0$ have a common root then the value of $|a|$ is

(A) 1 (B) 0 (C) -1 (D) None of these

[Ans. : A]

DIBY-56

If a, b, p, q are non-zero real numbers, then the equations $2a^2x^2 - 2abx + b^2 = 0$ and $p^2x^2 + 2pqx + q^2 = 0$ have:

(A) No common root

(B) One common root if $2a^2 + b^2 = p^2 + q^2$

(C) Two common roots if $3pq = 2ab$

(D) Two common roots if $3qb = 2ap$

[Ans. : A]

DIBY-57

$x^2 - 11x + a$ and $x^2 - 14x + 2a$ will have a common factor, if $a =$

(A) 24 (B) 0, 24 (C) 3, 24 (D) 0, 3

[Ans. : B]

DIBY-58

If $(x^2 + px + 1)$ is a factor of $(ax^3 + bx + c)$, then

(A) $a^2 + c^2 = -ab$ (B) $a^2 - c^2 = -ab$

(C) $a^2 - c^2 = ab$ (D) None of these

[Ans. : C]

DIBY-59

The value of p and q for which the equation $px^2 + 5x + 2 = 0$ and $3x^2 + 10x + q = 0$ have both roots is common.

(A) $p = \frac{3}{2}, q = 4$ **(B)** $p = 2, q = \frac{3}{4}$ **(C)** $p = \frac{2}{3}, q = 4$ **(D)** $p = 4, q = \frac{3}{2}$

[Ans. : (A)]

DIBY-60

If the equation $x^2 - 4x + 5 = 0$ and $x^2 + ax + b = 0$ (where $a, b \in \mathbb{R}$) have a common roots then which of the following is correct?

(A) $a = -4, b = -5$
(B) $a + b = 1$
(C) $a + b = -1$
(D) $b - a = 1$

[Ans. : (B)]

DIBY-61

If $Q_1(x) = x^2 + (k - 29)x - k$ and $Q_2(x) = 2x^2 + (2k - 43)x + k$ both are factors of a cubic polynomial $P(x)$, then the largest value of k is

(A) 0 **(B)** 33 **(C)** 23 **(D)** 30

[Ans. : (D)]

DIBY-62

If the equation $x^2 + abx + c = 0$ and $x^2 + acx + b = 0, (b \neq c)$ have a common root then their uncommon roots are the roots of the equation

(A) $a(b+c)x^2 + (b+c)x + abc = 0$ **(B)** $a(b+c)x^2 - (b+c)x - abc = 0$

(C) $a(b+c)x^2 - (b+c)x + abc = 0$ **(D)** $a(b+c)x^2 + (b+c)x - abc = 0$

[Ans. : (D)]

DIBY-63

If $(a-b)x^2 + 2(b-c)x + (a-c+1) = 0 (a, b, c \in \mathbb{R})$ and $x^2 + 2x + 5 = 0$ have a common roots, then which of the following is true?

(A) $a - b = 1$ **(B)** $a - b = \frac{1}{3}$ **(C)** $a - b = \frac{1}{2}$ **(D)** $a - b = 2$

[Ans. : (B)]

DIBY-64

If equations $ax^2 - (a+b)x + b = 0$ & $bx^2 + (b-c)x - c = 0$ has exactly one root in common $\{a, b, c \neq 0\}$, then which of the following can be correct?

(A) $b^2 = ac$ **(B)** $-a = b \neq c$ **(C)** $b = a \neq c$ **(D)** $-a \neq b = c$

[Ans. : A, B, D]

DIBY-65

Let $\lambda \neq 0$ be in R. If α and β are the roots of the equation, $x^2 - x + 2\lambda = 0$ and α and γ are the roots of the equation, $3x^2 - 10x + 27\lambda = 0$, then $\frac{\beta\gamma}{\lambda}$ is equal to:

[JEE MAINS - 2020]

(A) 18 (B) 36 (C) 9 (D) 27

[Ans. : (A)]

DIBY-66

If one root is square of the other root of equation $x^2 + px + q = 0$, then the relation between p and q is

[IIT-JEE-2004]

(A) $p^3 - q(3p - 1) + q^2 = 0$

(B) $p^3 - q(3p + 1) + q^2 = 0$

(C) $p^3 + q(3p - 1) + q^2 = 0$

(D) $p^3 + q(3p + 1) + q^2 = 0$

[Ans. : (A)]

LOCATION OF ROOTS

DIBY-67

Let $x^2-(m-3)x+m=0,(m\in R)$ be a quadratic equation. Find the value of m for which

(i) both the roots are smaller than 2.

(ii) both the roots are greater than 2.

(iii) one root is smaller than 2 and the other root is greater than 2.

(iv) exactly one root lies in the interval (1, 2).

(v) both the roots lie in the interval (1, 2).

(vi) one root is greater than 2 and the other root is smaller than 1.

(vii) atleast one root lie in interval (1, 2).

(viii) atleast one root is greater than 2.

Ans. :

(i) $m\in(-\infty,1]$, (ii) $m\in[9,10)$, (iii) $m\in(10,\infty)$, (iv) $m\in(10,\infty)$,
(v) $m\in\phi$, (vi) $m\in\phi$, (vii) $m\in(10,\infty)$, (viii) $m\in[9,\infty)-\{10\}$]

LOCATION OF ROOTS

DIBY-68

Find all values of a for which the equation $4x^2 - 2x + a = 0$ has two roots lie in the interval (–1, 1).

[Ans. : $a \in \left(-2, \frac{1}{4}\right]$]

DIBY-69

The number of integral values of k, for which one root of the equation $2x^2 - 8x + k = 0$ lies in the interval (1, 2) and its other root lies in the interval (2, 3), is **[JEE Mains-2023]**

(A) 2 (B) 0 (C) 1 (D) 3

[Ans. (C)]

DIBY-70

Find all the values of 'a' for which both roots of the equation $x^2 + x + a = 0$ exceed the quantity 'a'.

(A) $(-\infty, -2)$ (B) $\left(-\infty, \frac{1}{4}\right]$ (C) $(-2, 0)$ (D) $(-\infty, -2) \cup \left(0, \frac{1}{4}\right)$

[Ans. (A)]

LOCATION OF ROOTS

DIBY-71

Find the set of values of 'a' for which zeroes of the quadratic polynomial $(a^2+a+1)x^2+(a-1)x+a^2$ are located on either side of 3.

(A) $(-\infty, -2)$ **(B)** $(0, \infty)$ **(C)** ϕ **(D)** $\mathbb{R}$

[Ans. : (C)]

DIBY-72

Find all possible values of 'a' for which exactly one root of the quadratic equation $x^2-(a+1)x+2a=0$ lie in the interval (0, 3).

(A) $(-\infty, 0) \cup (6, \infty)$

(B) $(-\infty, 0] \cup (6, \infty)$

(C) $(-\infty, 0) \cup [6, \infty)$

(D) $(-\infty, 0] \cup [6, \infty)$

[Ans. : (B)]

HIGHER DEGREE POLYNOMIALS

DIBY-73

If α, β, γ are roots of $5x^3 - qx - 1 = 0, (q \in R)$ find the value of $\frac{\alpha^2-3}{\beta\gamma} + \frac{\beta^2-3}{\alpha\gamma} + \frac{\gamma^2-3}{\alpha\beta}$.

[Ans. : 3]

DIBY-74

Given that α, β, γ are the roots of cubic equation $x^3 - 3x^2 + 2x + \frac{2}{3} = 0$. The value of $\alpha^4 + \beta^4 + \gamma^4$ is equal to

[Ans. : 9]

DIBY-75

$P(x) = x^3 + 33x^2 + 327x + 935$. Let $P(x)$ be a polynomial as described above with $a; b; c$ the roots of $P(x)$. Find $a^2 + b^2 + c^2$ without solving $P(x) = 0$.

[Ans. : 435]

HIGHER DEGREE POLYNOMIALS

DIBY-76

Let α, β and γ are the roots of the cubic $x^3 - 3x^2 + 1 = 0$. Find a cubic whose roots are $\frac{\alpha}{\alpha-2}, \frac{\beta}{\beta-2}$ and $\frac{\gamma}{\gamma-2}$. Hence or otherwise find the value of $(\alpha-2)(\beta-2)(\gamma-2)$.

[Ans. : $3y^3 - 9y^2 - 3y + 1 = 0, (\alpha-2)(\beta-2)(\gamma-2) = 3$]

DIBY-77

If α is a root of $x^3 - x - 1 = 0$, then the value of $\alpha^{10} + 2\alpha^8 - \alpha^7 - 3\alpha^6 - 3\alpha^5 + 4\alpha^4 + 2\alpha^3 - 4\alpha^2 - 6\alpha + 2$ is

[Ans. : 4]

DIBY-78

Let α, β, γ be the roots of the cubic equation $x^3 - px^2 + qx - r = 0$. Find the value of sum of the cubes of the roots

(A) $p^3 - 3pq + 3r$

(B) $p^3 - 3pq - 3r$

(C) $p^3 + 3pq - 3r$

(D) $p^3 - 2pq + r$

[Ans. : (A)]

HIGHER DEGREE POLYNOMIALS

DIBY-79

If α, β, γ be the roots of the equation $x^3 + ax + a = 0$ (where $a \in \mathbb{R}, a \neq 0$) satisfy $\frac{\alpha^2}{\beta} + \frac{\beta^2}{\gamma} + \frac{\gamma^2}{\alpha} = -8$, then a is –

(A) –8 (B) –2 (C) 2 (D) 8

[Ans. : (A)]

DIBY-80

If a, b, c are the roots of cubic $x^3 - x^2 + 1 = 0$ then find the value of $a^{-2} + b^{-2} + c^{-2}$

(A) 0 (B) 1 (C) 2 (D) 3

[Ans. : (C)]

DIBY-81

If α, β, γ & δ are roots of equation $x^4 - 7x^2 + x - 5 = 0$, then value of $(\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\beta + \gamma + \delta)(\alpha + \gamma + \delta)$ is–

(A) –5 (B) 5 (C) 1 (D) –1

[Ans. : (A)]

HIGHER DEGREE POLYNOMIALS

DIBY-82

If α, β, γ are the roots of equation $x^3 + 2x - 5 = 0$ and if equation $x^3 + bx^2 + cx + d = 0$ has roots $2\alpha + 1, 2\beta + 1, 2\gamma + 1$, then value of $|b + c + d|$ is (where b, c, d are coprime)

(A) 41 (B) 39 (C) 40 (D) 43

[Ans. : (A)]

DIBY-83

The length of the sides of a triangle are the 3 distinct roots of the equation $4x^3 - 24x^2 + 47x - 30 = 0$. If the area of triangle is Δ, find the value of 100Δ.

(A) 100 (B) 200 (C) 150 (D) 250

[Ans. : (C)]

DIBY-84

Let for $a \neq a_1 \neq 0$, and $f(x) = ax^2 + bx + c, g(x) = a_1x^2 + b_1x + c_1$ and $p(x) = f(x) - g(x)$. If $p(x) = 0$ only for $x = -1$ and $p(-2) = 2$, then the value of $p(2)$ is

[AIEEE-2011]

(A) 18 (B) 3 (C) 9 (D) 6

[Ans. : (A)]

It is never about end result

It's always about Journey

THANK You

#futureIITians